

---

# Methods of Calculation and Design for Mechanical and Public Health Building Services

---

How-To: Mechanical and Public Health Building Services  
System Design

*Author:*

Adam Rees

# Mechanical Building Services Calculation Methods

By

Adam Rees

©Adam Rees, 2019

# Contents

|                                                  |           |
|--------------------------------------------------|-----------|
| <b>Acronyms</b>                                  | <b>1</b>  |
| <b>I Ventilation</b>                             | <b>3</b>  |
| <b>1 Introduction</b>                            | <b>4</b>  |
| 1.1 Energy Balances . . . . .                    | 4         |
| 1.2 Units . . . . .                              | 5         |
| 1.3 Shadow Gaps and Door Undercuts . . . . .     | 7         |
| 1.4 Ventilation Rates . . . . .                  | 8         |
| 1.4.1 Pollutant Control . . . . .                | 8         |
| 1.4.2 Kitchen Ventilation Rates . . . . .        | 9         |
| 1.4.2.1 Extract Sizing - Method 1 . . . . .      | 9         |
| 1.4.2.2 Extract Sizing - Method 2 . . . . .      | 9         |
| 1.4.2.3 Extract Sizing - Method 3 . . . . .      | 10        |
| 1.4.2.4 Extract Sizing - Method 4 . . . . .      | 10        |
| 1.4.2.5 Extract Sizing - Method 5 . . . . .      | 10        |
| <b>2 Psychrometrics</b>                          | <b>11</b> |
| 2.1 Psychrometric Charts . . . . .               | 11        |
| 2.1.1 Psychrometric Processes . . . . .          | 11        |
| 2.1.2 ADP Calculations . . . . .                 | 11        |
| 2.1.3 Room Condition Calculations . . . . .      | 11        |
| 2.1.4 Enthalpy Calculations . . . . .            | 11        |
| 2.2 Manual Psychrometric Calculations . . . . .  | 11        |
| <b>3 Heating and Cooling via Air</b>             | <b>12</b> |
| 3.1 Comfort and Efficiency . . . . .             | 12        |
| 3.1.1 Guides and Regulations . . . . .           | 12        |
| 3.2 Calculations . . . . .                       | 13        |
| 3.2.1 Heating by Air . . . . .                   | 13        |
| 3.2.2 Cooling by Air . . . . .                   | 13        |
| <b>4 Ventilation Plant</b>                       | <b>14</b> |
| 4.1 Air Handling Unit (AHU) Components . . . . . | 14        |
| 4.1.1 Energy Recovery Devices . . . . .          | 14        |

|           |                                                                          |           |
|-----------|--------------------------------------------------------------------------|-----------|
| 4.1.1.1   | Plate Heat Exchanger . . . . .                                           | 15        |
| 4.1.1.2   | Thermal Wheel . . . . .                                                  | 15        |
| 4.1.1.3   | Run-Around-Coil . . . . .                                                | 15        |
| 4.1.1.4   | Mixing Box . . . . .                                                     | 15        |
| 4.1.1.5   | ErP Regulations . . . . .                                                | 16        |
| 4.1.1.6   | Health Technical Memorandum (HTM) 03-01 Standard AHU<br>Layout . . . . . | 16        |
| 4.1.2     | Energy Recovery Calculations . . . . .                                   | 16        |
| 4.1.3     | Frost Coils . . . . .                                                    | 18        |
| 4.1.3.1   | Fog Coils . . . . .                                                      | 19        |
| 4.1.3.2   | Calculations with and without Frost Coils . . . . .                      | 19        |
| 4.1.4     | Filtration . . . . .                                                     | 21        |
| 4.1.5     | Fans . . . . .                                                           | 21        |
| 4.1.6     | Heating/Cooling Coils . . . . .                                          | 21        |
| 4.1.6.1   | Dehumidification . . . . .                                               | 21        |
| 4.2       | Component Resistance . . . . .                                           | 21        |
| 4.3       | Fan Heat Gain . . . . .                                                  | 21        |
| 4.4       | Ventilation Plant Running Costs . . . . .                                | 21        |
| <b>5</b>  | <b>Air Distribution</b>                                                  | <b>22</b> |
| 5.1       | Displacement Ventilation . . . . .                                       | 22        |
| 5.2       | Ductwork Component Pressure Loss . . . . .                               | 22        |
| 5.3       | Fire Dampers . . . . .                                                   | 22        |
| 5.3.1     | Fire Damper Types . . . . .                                              | 22        |
| 5.3.1.1   | Thermal Link Fire Dampers . . . . .                                      | 22        |
| 5.3.1.2   | Motorised Fire and Smoke Dampers . . . . .                               | 22        |
| 5.3.2     | Fire Damper Selection . . . . .                                          | 22        |
| <b>II</b> | <b>Heating and Cooling</b>                                               | <b>23</b> |
| <b>6</b>  | <b>Introduction</b>                                                      | <b>24</b> |
| 6.1       | Steady State Heating . . . . .                                           | 24        |
| 6.1.1     | Conduction (Fabric) Heat Loss . . . . .                                  | 24        |
| 6.1.2     | Infiltration Heat Loss . . . . .                                         | 25        |
| 6.1.3     | Ventilation Heat Loss . . . . .                                          | 25        |
| 6.1.4     | Intermittent Correction Factor . . . . .                                 | 25        |
| 6.1.5     | Peak Heating Load . . . . .                                              | 25        |
| 6.1.6     | Distribution Losses . . . . .                                            | 25        |
| 6.1.7     | Infiltration Diversity Ratio . . . . .                                   | 25        |
| 6.1.8     | Central Plant Size . . . . .                                             | 25        |
| 6.2       | Annual Estimation of Heating Loads . . . . .                             | 25        |

|                                                 |           |
|-------------------------------------------------|-----------|
| <b>III Domestic Water Services</b>              | <b>26</b> |
| <b>7 Water Tanks</b>                            | <b>27</b> |
| 7.1 Tank Types . . . . .                        | 27        |
| 7.1.1 Sectional Tanks . . . . .                 | 27        |
| 7.1.1.1 Externally Flanged . . . . .            | 27        |
| 7.1.1.2 Internally Flanged . . . . .            | 27        |
| 7.1.2 One piece GRP tanks . . . . .             | 27        |
| 7.1.3 Underground Tanks . . . . .               | 27        |
| 7.1.4 Cold Water Service (CWS) Tanks . . . . .  | 27        |
| 7.1.5 Hot Water Service (HWS) Tanks . . . . .   | 27        |
| 7.1.6 Sprinkler Tanks . . . . .                 | 27        |
| 7.2 Tank Sizing . . . . .                       | 28        |
| 7.2.1 Tanks in Small Spaces . . . . .           | 28        |
| 7.2.1.1 Access Space . . . . .                  | 28        |
| 7.2.1.2 Raised Ball Valve Chamber . . . . .     | 28        |
| 7.2.1.3 Manway . . . . .                        | 28        |
| 7.3 Tank Inlet . . . . .                        | 29        |
| 7.4 Gravity Discharge from a tank . . . . .     | 29        |
| 7.5 Solar Hot Water Tank Arrangment . . . . .   | 30        |
| 7.6 Feed and Expansion Tank . . . . .           | 30        |
| <b>8 Domestic Hot Water Loads</b>               | <b>31</b> |
| 8.1 Peak Loads . . . . .                        | 31        |
| 8.2 Estimating Annual Hot Water Loads . . . . . | 31        |
| <b>IV Drainage and Rainwater</b>                | <b>32</b> |
| <b>Bibliography</b>                             | <b>33</b> |

# Acronyms

**ACH** Air Changes per Hour.

**AHU** Air Handling Unit.

**ASHRAE** American Society of Heating, Refrigerating and Air-Conditioning Engineers.

**BB101** Building Bulletin 101 - Guidelines on ventilation thermal comfort and indoor air quality in schools.

**CIBSE** Chartered Institution of Building Services Engineers.

**CWS** Cold Water Service.

**HBN** Health Building Note.

**HTM** Health Technical Memorandum.

**HWS** Hot Water Service.

**VOC** Volatile Organic Compounds.

# To-Do:

|                                                                                                                                                                                                                                                        |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Review this section . . . . .                                                                                                                                                                                                                          | 8  |
| Figure out how to draw psychrometric processes in $\text{\LaTeX}$ . . . . .                                                                                                                                                                            | 11 |
| Include a schematic of the HTM 03-01 standard AHU layout, and reference it in the<br>text above. Check layout against the standard, and ensure that the component<br>numbering is consistent with the typical AHU schematic shown in Figure 4.1. . . . | 16 |
| Filter Resistance dirty/average/clean etc., coil resistance, PHX vs thermal wheel resistance                                                                                                                                                           | 21 |
| Put an intro to the tank section here. Also need to decide what is going in this section .                                                                                                                                                             | 27 |
| See Domestic Water System Calculator - Excel Sheet . . . . .                                                                                                                                                                                           | 28 |
| Include filtering (inc. automatic backwash filter), and valves . . . . .                                                                                                                                                                               | 29 |
| See notes in OneNote link . . . . .                                                                                                                                                                                                                    | 30 |

## Part I

# Ventilation

# Chapter 1

## Introduction

This chapter will outline some of the basics of ventilation system design, such as Units, and Ventilation Rates.

### 1.1 Energy Balances

When performing design calculations on air flow rates, we need to know the energy transfer from different fluids either through coils, mixed air, or other means. The following equations describe the energy balance commonly used within Building Services design.

Equation 1.1, shows the relationship between Energy in Kilo-Watts [kW], Mass Flow Rate in kilograms per second [kg/s], Specific Heat Capacity in Joules/Gram-Kelvin [JK/g], and Temperature Difference in Kelvin [K]. Equations 1.2 through 1.4, shows the dimensional analysis of equation 1.1.

$$Q = m C_p \delta T \quad (1.1)$$

Expressing equation 1.1 as units:

$$\left[ \frac{kg}{s} \right] \left[ \frac{J}{g \cdot K} \right] \left[ \frac{K}{1} \right] \quad (1.2)$$

Equation 1.2 can be reduced by removing Kelvin and grams from both the denominator and numerator:

$$\left[ \frac{kg}{s} \right] \left[ \frac{J}{g \cdot K} \right] \left[ \frac{K}{1} \right] \quad (1.3)$$

This gives equation 1.4, note the definition of Watts [W] as a Joule per second [J/s]:

$$\left[ \frac{1000}{s} \right] \left[ \frac{J}{1} \right] = \left[ \frac{1000}{1} \right] \left[ \frac{W}{1} \right] = \left[ \frac{kW}{1} \right] \quad (1.4)$$

See Equation 1.5 if you want to use volumetric flow rate [m<sup>3</sup>/s, or L/s] instead of mass flow rate [kg/s].

## 1.2 Units

Cubic metres per second [m<sup>3</sup>/s], litres per second [L/s], kilograms per second [kg/s], Air Changes per Hour (ACH) and so on.

- m<sup>3</sup>/s, is used for larger volumetric air flow rates, there are a 1,000 litres in a cubic meter of air.
- L/s, is used for smaller volumetric air flow rates (Single terminal units/grilles etc), a litre is defined as one cubic decimeter.
- kg/s, is used to describe the mass flow rate of air, and is used when calculating energy balances .
- ACH, is used to describe the turnover of air in a room.

Converting from volumetric flow [m<sup>3</sup>/s], [L/s] to mass flow [kg/s] requires the density of air [ $\rho$ ] at the working temperature. Table 1.1 shows the density of air at some common temperatures.

| Air Temperature<br>°C | Air Density ( $\rho$ )<br>kg/m <sup>3</sup> |
|-----------------------|---------------------------------------------|
| 0                     | 1.292                                       |
| 10                    | 1.246                                       |
| 20                    | 1.204                                       |
| 30                    | 1.164                                       |
| 40                    | 1.127                                       |
| 50                    | 1.093                                       |

Table 1.1: Density of air at differing air temperatures

So by using a volumetric ventilation rate, which is commonly used to describe the outdoor air provision per person (See Section 1.4 for more information), we need to include density in Equation 1.1 so that we get sensible answers. Equation 1.1, becomes equation 1.5. (Dimensional analysis follows in Equations 1.6 through 1.8, and 1.9 through 1.11 for L/s).

$$Q = m \rho C_p \delta T \quad (1.5)$$

Following the reduction procedure in section 1.1, we can express equation 1.5 in units leaving equation 1.6:

$$\left[ \frac{m^3}{s} \right] \left[ \frac{kg}{m^3} \right] \left[ \frac{J}{g \cdot K} \right] \left[ \frac{K}{1} \right] \quad (1.6)$$

Now removing  $[m^3/s]$ ,  $[g]$ , and  $[K]$  from both the denominator and numerator of equation 1.6:

$$\left[ \frac{m^3}{s} \right] \left[ \frac{kg}{m^3} \right] \left[ \frac{J}{g \cdot K} \right] \left[ \frac{K}{1} \right] \quad (1.7)$$

Leaving equation 1.8, again note the definition of Watts  $[W]$ :

$$\left[ \frac{1}{s} \right] \left[ \frac{1000}{1} \right] \left[ \frac{J}{1} \right] = \left[ \frac{1000}{1} \right] \left[ \frac{W}{1} \right] = \left[ \frac{kW}{1} \right] \quad (1.8)$$

The same procedure can be followed for litres per second  $[L/s]$  instead of cubic metres per second  $[m^3/s]$ , noting the definition of  $[L/s]$  as a cubic decimeter, i.e.  $1 m^3 = 1,000 l$ :

$$\left[ \frac{l}{s} \right] \left[ \frac{kg}{m^3} \right] \left[ \frac{J}{g \cdot K} \right] \left[ \frac{K}{1} \right] \quad (1.9)$$

To remove both the litres from the first term numerator, and cubic meters from the second term denominator, we need to divide litres by 1,000, i.e. multiply by  $1/1000$ .

$$\left[ \frac{l}{s} \right] \left[ \frac{1}{1000} \right] \left[ \frac{kg}{m^3} \right] \left[ \frac{J}{g \cdot K} \right] \left[ \frac{K}{1} \right] \quad (1.10)$$

$$\left[ \frac{1}{s} \right] \left[ \frac{J}{1} \right] = \left[ \frac{W}{1} \right] \quad (1.11)$$

the primary difference between using  $[m^3/s]$  and  $[L/s]$  is the addition of  $1/1000$  when cancelling the  $[m^3]$  and  $[L]$ ; however, the  $1/1000$  further cancels out and the end result provides Watts instead of kilo-Watts.

### 1.3 Shadow Gaps and Door Undercuts

Door undercuts are used as a method of distributing air between rooms without ductwork, by simply allowing space beneath the door for air to pass then sufficient air can be introduced for a small occupied space provided that there is a method of extraction on the other side of the door. The supply and extraction provides the air flow between the rooms, and the door undercut provides the air path.

**Note:** *Door undercuts are not suitable for all applications, for example, if the door is a fire door, then the undercut may not be possible due to fire regulations. In this case, a transfer duct may be required to provide the necessary air flow between rooms which can be installed with a fire damper to maintain the integrity of the fire wall. See Section 5.3 for more information on fire and fire/smoke dampers.*

Shadow Gaps can provide an air path to a return air plenum, this avoids to use of return air grilles. This can provide a more aesthetically pleasing finish, but needs close coordination with the Architectural design and detailing to ensure that the gap is correctly sized and located.

The calculation of free area is the same for both door undercuts and shadow gaps. The difference being that the length of the door undercut is determined by the width of the door so only the height is adjustable, whereas the length and width of a shadow gap is adjustable. Equation 1.12, shows the calculation of free area for a given volumetric flow rate, where  $Q$  is the volumetric flow rate [ $\text{m}^3/\text{s}$ ],  $v$  is the velocity of the air stream through the gap [ $\text{m}/\text{s}$ ], and  $A$  is the free area [ $\text{m}^2$ ].

$$Q = vA \quad (1.12)$$

When using Equation 1.12 to calculate the free area of a door undercut, then by knowing the door width, the height of the undercut can be specified. This two step calculation can be combined into one as shown in equation 1.13, where  $h$  is the undercut height [ $\text{m}$ ], and  $w$  is the door width [ $\text{m}$ ].

$$h = \frac{Q}{wv} \quad (1.13)$$

Sensible air velocities for shadow gaps and door undercuts are below 2.0 m/s. This avoids excess pressure, and noise generation.

## 1.4 Ventilation Rates

The amount of air supplied or extracted from a space differs depending on the purpose of ventilation. If air is supplied to maintain a good indoor air quality then the concentration of pollutants (Such as CO<sub>2</sub>) in both the supply air and indoor air needs to be considered alongside the generation of pollutants, see subsection 1.4.1. Rules of thumb for the amount of outdoor air that should be provided per person are frequently given in guides such as Chartered Institution of Building Services Engineers (CIBSE) Guide A, some outdoor air rates per person and the expected quality of indoor air are given in Table 1.2.

| Supplied Air Rate<br>l/s/p | Indoor Air Quality | Classification<br>[BS EN 13779] | Total Indoor CO <sub>2</sub><br>[ppm] |
|----------------------------|--------------------|---------------------------------|---------------------------------------|
| 6                          | Low                | IDA4                            | 1600                                  |
| 8                          | Moderate           | IDA3                            | 1200                                  |
| 12.5                       | Medium             | IDA2                            | 900                                   |
| 20                         | High               | IDA1                            | 750                                   |

Table 1.2: Commonly Used Outdoor Air Rates, see CIBSE Guide A and BS EN 13779

### 1.4.1 Pollutant Control

Review this section

An internal space may suffer from a build-up of pollutants such as CO<sub>2</sub>, for Volatile Organic Compounds (VOC). The levels of these pollutants within the space can be controlled using mechanical ventilation, by providing external to the space and removing some of the internal air, the concentration of the pollutant can be reduced. Equations 1.14, and 1.15, describe the build-up of a pollutant both over time, and at a steady state condition.

CO<sub>2</sub> is typically released through respiration, an adult human would typically release 0.005 L/s each, and outdoor air typically contains 310 ppm of CO<sub>2</sub> (0.00031 L per L/s of outdoor air).

$$C_t = \frac{q_v c_0 + q_e}{q_v + q_e} \left[ 1 - e^{-\frac{q_v + q_e}{V} t} \right] \quad (1.14)$$

$$C_E = \frac{q_v c_0 + q_e}{q_v + q_e} \quad (1.15)$$

## 1.4.2 Kitchen Ventilation Rates

In the DW172 document published by the Heating and Ventilation Contractors Association (HVCA), there are five methods of determining the appropriate extract rate for kitchen extract systems. A mechanical consultant will usually have the extract rate specified by the kitchen specialist, but early stage designs specialists are not usually appointed so an estimation is usually required.

If mechanical supply air is to be provided to the space it is done so at 85% of the extract volumetric flow rate.

The thermal convection method (Method 1) is the only acceptable method for final design, the other four methods should only be used for indicative or provisional extract flow rates.

### 1.4.2.1 Extract Sizing - Method 1

This method uses the thermal convection method, and specifies a  $[m^3/s]$  per  $[m^2]$  of catering appliance work area. Table 2 in DW172 shows volumetric extract flow rates for different appliances, then applying a canopy factor. An example of calculating the total extract rate is given in Table 1.3.

| Item                                                   | Area   | Fuel     | Coefficient | Flow Rate     |
|--------------------------------------------------------|--------|----------|-------------|---------------|
| Griddle                                                | 0.4500 | Gas      | 0.30        | 0.135         |
| Open Top Range                                         | 0.6750 | Gas      | 0.35        | 0.236         |
| Solid Top Range                                        | 0.5625 | Gas      | 0.60        | 0.338         |
| Bench                                                  | 0.3750 | -        | 0.03        | 0.011         |
| Twin Fryers                                            | 0.4875 | Electric | 0.35        | 0.171         |
| Salamander Grille                                      | 0.3000 | Gas      | 0.75        | 0.225         |
| Sub-total Extract Volume                               |        |          |             | 1.116         |
| Canopy Factor - <i>Overhead wall open at both ends</i> |        |          |             | 1.25          |
| Volumetric Extract Rate Required                       |        |          |             | 1.395 $m^3/s$ |

Table 1.3: Example method of calculating the extract flow rate using Method 1 from DW172

### 1.4.2.2 Extract Sizing - Method 2

If the size of the canopy is known, but not the full detail of the catering equipment being installed, then by using the face velocity method the total extract rate can be estimated.

Table 1.4 shows the velocities for different equipment types, and the fastest face velocity should be used.

| Loading                       | Face Velocity |
|-------------------------------|---------------|
| Light (Boiling pans etc.)     | 0.25 m/s      |
| Medium (Deep fat fryers etc.) | 0.35 m/s      |
| Heavy (Chargrills etc.)       | 0.5 m/s       |

Table 1.4: Face velocity for the calculation of the extract flow rate using Method 2 from DW172

#### 1.4.2.3 Extract Sizing - Method 3

A volumetric extract flow rate calculated by assigning a  $[\text{m}^3/\text{s}]$  value per  $[\text{kW}]$  of power input, the total air supply is then totalled.

#### 1.4.2.4 Extract Sizing - Method 4

Method 4 is based on ACH, starting at 40 ACH for volumetric extract and increasing this if high output or densely packed equipment is used. ACH rates of up to 120 can be seen, and 40 ACH is regarded as the minimum.

#### 1.4.2.5 Extract Sizing - Method 5

An extract flow rate is assigned per metre length of extract hood, a table in DW172 has been provided that gives extract rates for different canopy types and duty levels.

## Chapter 2

# Psychrometrics

Figure out how to draw psychrometric processes in L<sup>A</sup>T<sub>E</sub>X

Use my [OneNote Psychrometric Notes](#) to help here!

### 2.1 Psychrometric Charts

#### 2.1.1 Psychrometric Processes

#### 2.1.2 ADP Calculations

#### 2.1.3 Room Condition Calculations

#### 2.1.4 Enthalpy Calculations

### 2.2 Manual Psychrometric Calculations

## Chapter 3

# Heating and Cooling via Air

General heating and cooling requirements and calculations are covered in Section 6. The specific requirements and calculations for heating and cooling spaces via air systems are covered in this chapter.

Once the heating and cooling loads are calculated, the next step is to determine the supply air temperature and flow rate required to meet those loads.

### 3.1 Comfort and Efficiency

- Supply air temperatures,
- Air flow rates and velocities,
- Noise levels,
- Humidity control,
- Energy efficiency.

#### 3.1.1 Guides and Regulations

- Health Technical Memorandum (HTM)
- Health Building Note (HBN)
- Building Regulations,
- Building Bulletin 101 - Guidelines on ventilation thermal comfort and indoor air quality in schools (BB101),

- CIBSE Guides,
- American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Standards.

## 3.2 Calculations

### 3.2.1 Heating by Air

$$Q = m C_p \delta T$$

### 3.2.2 Cooling by Air

$$Q = m \delta H$$

## Chapter 4

# Ventilation Plant

### 4.1 Air Handling Unit (AHU) Components

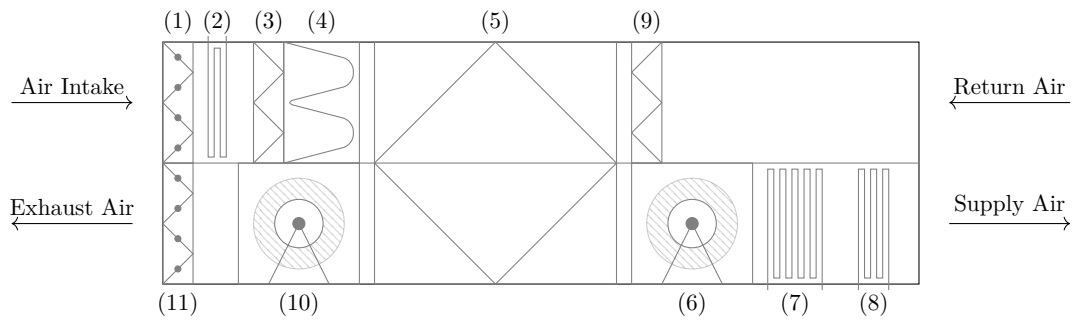


Figure 4.1: Typical AHU schematic, the component numbering follows the air path from outdoor air (Air Intake) through to the space (Supply Air), and then from the return air back to exhaust air.

A typical AHU schematic is shown in figure 4.1, (1) & (11) are motorised dampers, (2) is a frost coil, (3) & (9) are panel filters, (4) is a bag filter, (5) is a counter-flow plate heat exchanger, (6) & (10) are supply and extract fans respectively, (7) is a cooling coil, (8) is a heating or re-heat coil.

#### 4.1.1 Energy Recovery Devices

A number of heat recovery devices are available for inclusion within an AHU.

- Plate Heat Exchanger,
- Thermal Wheel,

- Run-Around-Coil,
- Mixing Box,

#### **4.1.1.1 Plate Heat Exchanger**

The plate heat exchanger works by passing the opposing air streams (supply and extract) through fins within the unit and heat transfer occurs through the metal fins without the two air streams coming into contact with each other. See Figure 4.2 for a schematic example of a plate heat exchanger in a cross flow configuration.

#### **4.1.1.2 Thermal Wheel**

Thermal wheel units utilise a metal perforated wheel which is constantly rotating, this warms up in the extract air stream and then the warm section is rotated into the supply air stream and transfers heat to the air. Thermal wheels mix small amounts of the air streams, and involves the running of a motor to drive the wheel.

#### **4.1.1.3 Run-Around-Coil**

Run-Around-Coil units use two finned coils (similar to the heating and cooling coils) filled with a glycol/water mixture, and run the mixture between the two coils positioned in each of the air streams (supply and extract). This system utilises a pump, and associated pressurisation equipment in order to pass water/glycol between the two coils.

#### **4.1.1.4 Mixing Box**

Mixing Box units recirculate a proportion of the air streams in order to satisfy external air supply and heating/cooling loads simultaneously, for example if the air supply to support the cooling load of a space is double the outdoor air provision, then a mixing box can be used to recirculate half the cooling load air flow rate and allow the rest to come from outside. This can reduce the amount of conditioning required for the supply air as the recirculated air may be closer to the required set-point than the external air.

#### 4.1.1.5 ErP Regulations

Under the ErP regulations<sup>1</sup>, some form of energy recovery must be used and only plate heat exchangers, thermal wheels, or run-around-coils are acceptable. In general the regulations apply to bi-directional and uni-directional ventilation units within commercial sized buildings; however, there are exceptions for certain applications such as Swimming Pools, Agricultural Buildings, Foundries etc.

#### 4.1.1.6 HTM 03-01 Standard AHU Layout

HTM 03-01 prescribes a standard AHU layout for healthcare applications, this layout includes a plate heat exchanger as the energy recovery device. The layout includes a fog coil, which is designed to prevent ice build-up on the filters and plate heat exchanger during the winter design condition, which is typically fully saturated air at temperatures lower than 0 °C. The layout also includes a cooling coil and a heating coil, which are used to condition the supply air to the required set point. The cooling coil is positioned before the heating coil to prevent any potential freezing within the coil; however, this precludes the ability to dehumidify the air stream should it be necessary/desired. The layout also includes a bag filter on the supply air side, and panel filters on the extract air side, this is to prevent any potential damage to the energy recovery device from dirty air, and to ensure that the supply air is clean for the space.

Include a schematic of the HTM 03-01 standard AHU layout, and reference it in the text above.

Check layout against the standard, and ensure that the component numbering is consistent with the typical AHU schematic shown in Figure 4.1.

### 4.1.2 Energy Recovery Calculations

Equation 4.1 shows the equation describing the efficiency of the heat recovery unit ( $\eta$ ) and the temperatures at each point for balanced air flow rates (i.e. supply air flow rate is equal to extract air flow rate), this equation is derived in Derivation 1 and shows allows for unbalanced air streams (i.e. supply air rate does not equal the extract air rate). Figure 4.2 shows a plate heat exchanger (PHX) and the appropriate labels to correspond to Equation 4.1. Minimum values prescribed in the ErP regulations for heat recovery efficiency can be found in Table 4.1.

$$\eta = \frac{t_2 - t_1}{t_3 - t_1} = \frac{t_{co} - t_{ci}}{t_{hi} - t_{ci}} \quad (4.1)$$

<sup>1</sup>Known by many different names: Energy related Products (ErP), Ecodesign Directive (Nº 2009/125/EC), Lot 6 - Ventilation Units, Ecodesign Regulation (Nº 1253/2014), UK Statutory Instrument 2015 (Nº 469)

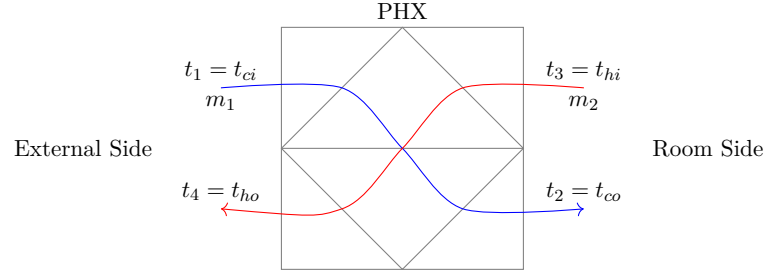


Figure 4.2: A counter-flow plate heat exchanger (PHX) showing air flow paths and temperature labels corresponding to Equation 4.1.

| Heat Recovery Type   | ErP 2018 Minimum Efficiency |
|----------------------|-----------------------------|
| Plate Heat Exchanger | 73%                         |
| Thermal Wheel        | 73%                         |
| Run-Around-Coil      | 68%                         |

Table 4.1: Table showing the minimum efficiency required for the three acceptable heat recovery equipment types in air handling units

**Derivation 1** *This is a derivation of the energy balance within an AHU heat recovery device allowing for unbalanced air flow rates. First we start with the basic energy balance equation defined in Equation 1.1.*

$$Q = m C_p \delta t$$

*Now we consider the supply air air-path (Cold), and the extract air air-path (Hot) separately.*

$$Q_{cold} = m C_p (t_{co} - t_{ci})$$

$$Q_{hot} = m C_p (t_{hi} - t_{ho})$$

*Using the above energy balances we can calculate the cold out temperature (Supply Air, Room side) if we know the temperature of the other three air streams, as this is rarely the case, we need to make some assumptions*

*If the heat exchanger were to be infinitely long, then the temperature of the hot in and cold out positions would be equal, i.e.*

$$t_{hi} = t_{co}$$

*Now in order to know the air temperature leaving the heat exchanger on the supply air side ( $t_{co}$ ), we would only have to know what the extract air temperature is ( $t_{hi}$ ); similarly, the temperature of the hot out ( $t_{ho}$ ) and cold in ( $t_{ci}$ ) positions would be equal, i.e.*

$$t_{ho} = t_{ci}$$

*An infinitely long heat exchanger is impractical (and impossible), so we can account for the reduction in heat transfer by using an efficiency term ( $\eta$ ) as a ratio of the energy balances, i.e.*

$$\frac{Q_{1,cold}}{Q_{2,hot}} = \frac{m_1 C_p (t_{co} - t_{ci})}{m_2 C_p (t_{hi} - t_{ho})}$$

let  $\eta \times t_{ci} = t_{ho}$

$$\eta = \frac{m_1 \cancel{C_p} (t_{co} - t_{ci})}{m_2 \cancel{C_p} (t_{hi} - t_{ci})}$$

using the labels in Figure 4.2, the equation becomes;

$$\eta = \frac{m_1 t_2 - t_1}{m_2 t_3 - t_1}$$

***QED***

### 4.1.3 Frost Coils

During the winter ventilation design condition, which is typically fully saturated air at temperatures lower than 0 °C, the build-up of ice within an AHU is possible. Ice can build-up on filters, plate-heat exchangers, or coils that are not heated (i.e. cooling coils or in-active heating coils), but typically only when the unit is running. To combat the ice build up, frost coils can be installed which are sized with sufficient capacity to raise the temperature from the external design condition (say −5 °C) to a positive temperature (typically 5 °C).

Whilst the unit is operational it is fair to assume that the heating coil and energy recovery devices will be activated and so any damage from ice build up will be limited to the filters (shown at locations (2) and (3) in Figure 4.1), as the energy recovery device will warm the incoming air using the return air. When a filter is frozen there is a risk that the filter will break up, or be pulled through the unit by the action of the fans, which will potentially damage the energy recovery device, or downstream filters. Certain manufacturers will guarantee that their filters can be used in freezing conditions without breaking, therefore negating the need for a frost coil but bringing a component of risk.

Using a frost coil within an AHU will increase the internal static pressure required to be overcome by the fans (Increasing specific fan power (SFP)), reduce the difference in temperature across the energy recovery device (reducing the 'free' energy applied to the supply air stream<sup>2</sup>), and increase the cost of both the unit and installation.

If a frost coil is omitted from an AHU then additional precautions are required, for example, additional control routines would be built into the control system that would shut dampers if the unit begins to freeze and turn the heating coil to full to attempt to defrost any frozen components. If a cooling coil is present within the unit, then it would need to be sited after

---

<sup>2</sup>'free' in the sense of recovered energy that would otherwise be lost to atmosphere without the energy recovery device

the heating coil to prevent any potential freezing within the coil; however, this precludes the ability to dehumidify the air stream should it be necessary/desired.

#### 4.1.3.1 Fog Coils

The concept of a fog coil was introduced by HTM 03-01<sup>3</sup> as an alternative to the standard frost coils described previously in subsection 4.1.3. The fog coil is designed to increase the incoming temperature by a fixed amount, typically, 3 °C, this moves the incoming air condition away from the dew point without the need to raise the temperature to a positive value, and therefore reduces the energy demand of the coil and the associated increase in fan power.

#### 4.1.3.2 Calculations with and without Frost Coils

If we consider two AHU with both with a design volumetric air flow rate of 1 m<sup>3</sup>/s, a energy recovery device with an efficiency,  $\eta$ , of 73%, and one unit with a frost coil and one without, we can compare the difference in running costs. We are considering the winter time design, so external temperature is −5 °C, and the AHU supply temperature set point is 20 °C.

Lets consider the LTHW load, Equation 4.5 shows the required LTHW mass flow rate with a frost coil at our design duty, and Equation 4.8 shows the required LTHW mass flow rate without a frost coil.

Firstly the AHU with a frost coil, we consider the energy required to heat 1 m<sup>3</sup>/s of air from −5 °C to 5 °C:

$$Q = 1 \times 1.2 \times 1.005 \times (5 - -5) = 12.06kW \quad (4.2)$$

Now with a off coil temperature of 5 °C from our frost coil and a return air temperature of 20 °C we can calculate the temperature gain across our energy recovery device (Using Equation 4.1 described earlier):

$$t_2 = (0.73 \times (20 - 5)) + 5 = 15.95^\circ\text{C}, \text{ say } 16^\circ\text{C} \quad (4.3)$$

Finally we need the energy required to heat our air stream from 16 °C to 20 °C:

$$Q = 1 \times 1.2 \times 1.005 \times (20 - 16) = 4.82kW \quad (4.4)$$

---

<sup>3</sup>HTM 03-01: Specialised ventilation for healthcare premises, Part A: Design and validation, 2018

So in total we require  $12.06 + 4.82 = 16.88kW$  of LTHW input, the mass flow rate of which is:

$$m = \frac{16.88}{4.187 \times (80 - 60)} = 0.20kg/s \quad (4.5)$$

Now we consider the same AHU without the frost coil, ignoring any detriment to frozen filters etc. We know that the air stream is at  $-5^\circ C$  and our return air is at  $20^\circ C$ , so the temperature rise across our energy recovery device is:

$$t_2 = (0.73 \times (20 - -5)) + -5 = 13.25^\circ C, \text{ say } 13^\circ C \quad (4.6)$$

This is only three degrees lower than our example with a frost coil, meaning that we recover a lot more heat due to the larger temperature difference, raising the temperature  $18^\circ C$  compared to  $6^\circ C$  with a frost coil. Now we need to calculate the required energy to heat our air stream from  $13^\circ C$  to  $20^\circ C$ :

$$Q = 1 \times 1.2 \times 1.005 \times (20 - 13) = 8.44kW \quad (4.7)$$

The LTHW input to this unit is:

$$m = \frac{8.44}{4.187 \times (80 - 60)} = 0.10kg/s \quad (4.8)$$

This difference in LTHW mass flow rate is vast, and when considering LTHW generation via natural gas boilers (assuming 90% efficiency) the running cost is as below:

Firstly the unit with frost coils:

$$0.20/0.9 = 0.22kg/s \quad (4.9)$$

$$Q = 0.22 \times 4.187 \times (80 - 60) = 18.42kW \quad (4.10)$$

And assuming a cost of 2.57 p/kWh, the LTHW required by the AHU with frost coils costs £11.36 per 24 hours when the external temperature is  $-5^\circ C$ . The AHU without frost coils:

$$0.10/0.9 = 0.11kg/s \quad (4.11)$$

| Pre-Coil Type | HR Temperature (ho) | Heating Demand | Cost per 24 hours |
|---------------|---------------------|----------------|-------------------|
| Frost         | 16 °C               | 0.20 kg/s      | £11.36            |
| Fog           | 14 °C               | 0.13 kg/s      | £7.43             |
| None          | 13 °C               | 0.10 kg/s      | £5.68             |

Table 4.2: Table showing differences in AHU running costs both with and without a frost coil. The AHU considered has a flow rate of 1 m<sup>3</sup>/s, a energy recovery device with an efficiency of 73%, and a supply air set point of 20 °C. The external air temperature in this example is −5 °C and fully saturated.

$$Q = 0.11 \times 4.187 \times (80 - 60) = 9.21 kW \quad (4.12)$$

Using the costs above, the LTHW generation for the AHU without frost coils costs £5.68 per 24 hours. Although the cost is relatively small this is the cost of generating LTHW to serve the AHU demand, and the difference can add up when considering the complete winter period where temperatures can be negative for a lot longer than 24 hours! This cost analysis does not consider the increase in fan running cost due to the increased resistance from the frost coil, which affects the running cost year-round whether the frost coil is active or not. Table 4.2 give a summary of the running energy and costs for an AHU with and without a frost coil, a fog coil has also been included in the table for comparison.

#### 4.1.4 Filtration

#### 4.1.5 Fans

#### 4.1.6 Heating/Cooling Coils

##### 4.1.6.1 Dehumidification

### 4.2 Component Resistance

Filter Resistance dirty/average/clean etc., coil resistance, PHX vs thermal wheel resistance

### 4.3 Fan Heat Gain

### 4.4 Ventilation Plant Running Costs

## Chapter 5

# Air Distribution

### 5.1 Displacement Ventilation

### 5.2 Ductwork Component Pressure Loss

### 5.3 Fire Dampers

This section will cover the application and specification of fire dampers and fire/smoke dampers. The purpose of a fire damper is to prevent the spread of fire and/or smoke through the ductwork. Fire dampers are typically installed in the ductwork at the point where it penetrates a fire-rated wall or floor. They are designed to automatically close when exposed to high temperatures, thereby preventing the passage of fire and smoke.

*For the purposes of this section, fire damper references in headings will refer to both fire dampers and fire/smoke dampers, unless otherwise noted.*

#### 5.3.1 Fire Damper Types

##### 5.3.1.1 Thermal Link Fire Dampers

##### 5.3.1.2 Motorised Fire and Smoke Dampers

#### 5.3.2 Fire Damper Selection

## Part II

# Heating and Cooling

# Chapter 6

## Introduction

### 6.1 Steady State Heating

There are several steps to calculate the required heating for a room and the size of the central plant that is required, the general procedure is as follows:

- Calculate Conduction (Fabric) Heat Loss,
- Calculate Infiltration Heat Loss,
- Calculate Ventilation System Heat Loss (If Applicable),
- Allow for Local Effects,
- Determine Intermittent Correction Factor,
- Calculate Peak Heating Load,
- Size/Select Heat Emitters,
- Calculate Distribution System Heat Loss,
- Determine Infiltration Diversity Ratio,
- Calculate Central Plant Size,
- Select Central Plant, including any redundancy.

#### 6.1.1 Conduction (Fabric) Heat Loss

$$Q = UA \tag{6.1}$$

Define U-value here...

$$U = \frac{1}{(\sum R + R_i + R_o)} \quad (6.2)$$

Where R is the thermal resistance of the elements,  $R_i$  and  $R_o$  are the internal and external thermal resistances respectively.

[Designing Buildings Wiki on U-values](#)

### **6.1.2 Infiltration Heat Loss**

e.g. A air pressure test result of  $5 \text{ m}^3/(\text{m}^2 \text{ h})$  at 50 Pa would result in a air infiltration rate of 0.111 ACH.

### **6.1.3 Ventilation Heat Loss**

### **6.1.4 Intermittent Correction Factor**

### **6.1.5 Peak Heating Load**

### **6.1.6 Distribution Losses**

### **6.1.7 Infiltration Diversity Ratio**

### **6.1.8 Central Plant Size**

## **6.2 Annual Estimation of Heating Loads**

## **Part III**

# **Domestic Water Services**

# Chapter 7

## Water Tanks

Put an intro to the tank section here. Also need to decide what is going in this section

### 7.1 Tank Types

#### 7.1.1 Sectional Tanks

Sectional tanks usually require levelling steels as a base to be installed upon, these can be 50 mm high.

##### 7.1.1.1 Externally Flanged

##### 7.1.1.2 Internally Flanged

#### 7.1.2 One piece GRP tanks

#### 7.1.3 Underground Tanks

#### 7.1.4 Cold Water Service (CWS) Tanks

#### 7.1.5 Hot Water Service (HWS) Tanks

#### 7.1.6 Sprinkler Tanks

Sprinkler tanks may need to conform to LPCB requirements. In terms of sizing and specifying a tank, this means that a raised ball-valve chamber is required in order to maintain a class AB air gap.

## 7.2 Tank Sizing

See Domestic Water System Calculator - Excel Sheet

### 7.2.1 Tanks in Small Spaces

Sometimes a tank of a certain capacity is required but the space to install a tank is limited. The following subsections detail the usually allowed dimensions and sizes, alongside alternatives that may enable a tank to squeeze into a tight space.

#### 7.2.1.1 Access Space

Usually 600 mm is allowed as a minimum around the tank in order to tighten the bolts on an externally flanged tank; however, by using a totally internally flanged tank (described in sub-subsection 7.1.1.2), the access space can be reduced to nothing and the tank installed within the corner of a room. Access is still required to the manway and valve chambers.

#### 7.2.1.2 Raised Ball Valve Chamber

Typically a raised ball valve chamber can help with space restrictions where the room height is not the same throughout the room; for example, where the ceiling is stepped or sloped. The raised ball valve chamber can be positioned at the higher end of the room allowing the full height of the tank to be used to store water in the lower end of the room. Typical raised ball valve chambers are 450 mm high, whereas manufacturers sometimes offer smaller 'special' raised ball valve chambers such as a 250 mm high chamber from Balmoral Tanks which has side access to the valve.

#### 7.2.1.3 Manway

Access to the internals of the tank is required for periodic cleaning when the tank is drained. Standard arrangements involve manways located on the top surface of the tank, complete with a fixed 'cat' ladder. Where height is restricted, the manway can be relocated to the side wall of the tank, with a standard ladder on both the inside and outside of the tank to allow access, one drawback is that the tank has to be drained before the manway can be opened.

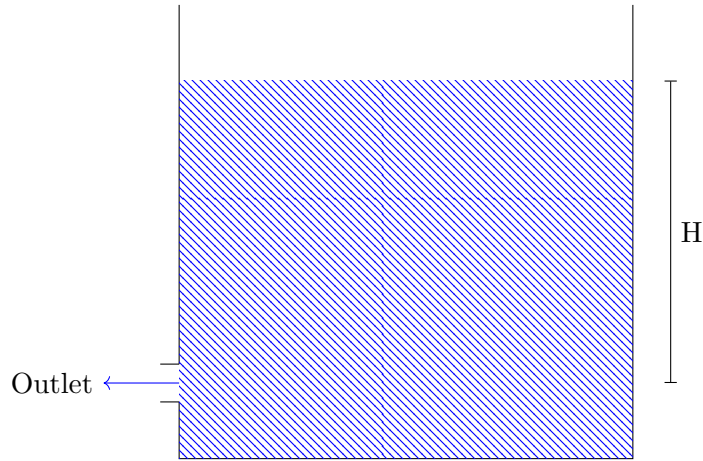


Figure 7.1: Sidewall gravity discharge from a tank

### 7.3 Tank Inlet

Include filtering (inc. automatic backwash filter), and valves

### 7.4 Gravity Discharge from a tank

Discharge from a tank by gravity alone is dependant on the following; Outlet Area ( $A$ ), Acceleration due to Gravity ( $g$ ), Height from the outlet to the water level ( $H$ ), and the Coefficient of Discharge ( $C_d$ ). Figure 7.1 shows a tank open to atmosphere with a side-wall discharge, and Equation 7.1 describes the volumetric flow rate through the outlet based on gravity flow, and an example tank discharge has been calculated in Equation 7.2.

$$Q = C_d A (2gH)^{\frac{1}{2}} \quad (7.1)$$

$$Q = 0.5 \times 0.00008 \times (2 \times 9.81 \times 0.8)^{0.5} = 0.00016 \text{ m}^3/\text{s} = 0.16 \text{ litre/s} \quad (7.2)$$

To calculate the centre line velocity of the discharge ( $U$ ), Equation 7.3 is used, and example for the same tank has been shown in Equation 7.4.  $C_v$  has been assumed to be 0.97 in this example.

$$U = C_v (2gH)^{\frac{1}{2}} \quad (7.3)$$

$$U = 0.97 \times (2 \times 9.81 \times 0.8)^{0.5} = 3.84 \text{ m/s} \quad (7.4)$$

Finally the reaction force ( $F$ ) from the water nozzle can be calculated from Equation 7.5, the density of water ( $\rho$ ) has been taken as  $1.000 \text{ kg/m}^3$ . The example tank has been shown in Equation 7.6

$$F = \rho QU \quad (7.5)$$

$$F = 1,000 \times 0.00016 \times 3.84 = 0.61 \text{ kg m/s}^2 = 0.61 \text{ N} \quad (7.6)$$

## 7.5 Solar Hot Water Tank Arrangement

See notes in OneNote link

[OneNote Solar HW Integration Notes](#)

## 7.6 Feed and Expansion Tank

## Chapter 8

# Domestic Hot Water Loads

### 8.1 Peak Loads

### 8.2 Estimating Annual Hot Water Loads

## Part IV

# Drainage and Rainwater

# Bibliography